

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

Anonymous authors
Anonymous submission

ABSTRACT

Representation alignment (REPA) accelerates generative training by distilling features from a pretrained vision encoder into intermediate diffusion representations. We ask which quantitative property of the target representation best predicts generation quality: its *global* semantic information, commonly measured by ImageNet-1K accuracy, or its *spatial* structure, measured by pairwise cosine similarity among patch tokens? Contrary to the prevailing expectation that stronger global semantic performance should provide the better target, our empirical comparisons across 27 vision encoders and multiple model scales show a consistent pattern: spatial structure is the stronger driver of REPA generation performance. We then test this evidence directly with two simple spatially focused modifications. We replace the standard MLP projection in REPA with a convolutional projection layer and apply spatial normalization to the external representation. The resulting method, iREPA, requires <4 lines of code yet consistently improves REPA convergence across vision encoders, model sizes, and training variants, including REPA, REPA-E, Meanflow, and JiT. These results reframe representation alignment around measurable spatial transfer and motivate target-representation selection criteria that emphasize practical generation performance rather than global accuracy alone.

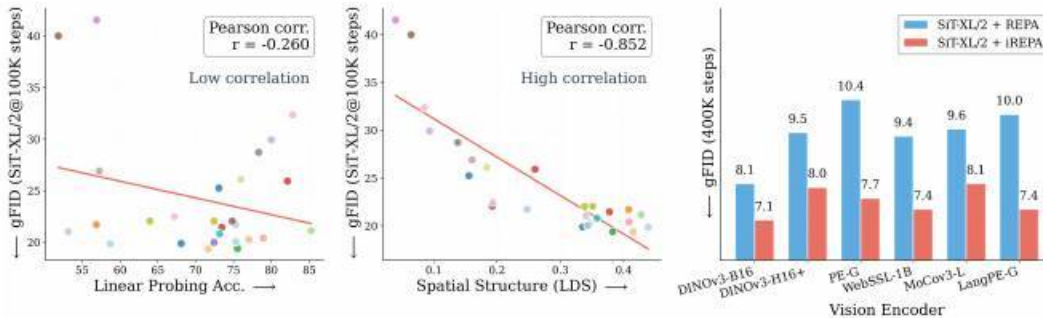


Figure 1: **What matters for representation alignment?** *Left:* Across 27 vision encoders, spatial structure provides a much stronger quantitative signal for generation performance than global performance measured by linear probing accuracy. *Right:* Building on this comparison, iREPA accentuates spatial transfer from the target representation to the diffusion model and consistently improves REPA convergence across diverse settings.

1 INTRODUCTION

Representation alignment is now a quantitatively effective way to accelerate diffusion-transformer training (Ma et al., 2024; Peebles & Xie, 2023). By matching intermediate diffusion features to pretrained self-supervised vision representations, REPA-style methods improve both convergence speed and final sample quality (Yu et al., 2024; Leng et al., 2025a). The empirical gains raise a more precise comparison question: when two target encoders are evaluated under the same REPA training and FID protocol, are gains better predicted by their global semantic information, typically summarized by ImageNet-1K linear probing accuracy, or by their spatial structure, measured through pairwise relationships among patch-token representations?



Figure 2: **Motivating examples — spatial structure matters.** Both panels compare ImageNet-1K accuracy against REPA generation quality under the same 100K-step SiT-XL/2 setting. *Left:* PE-G has much higher validation accuracy than SpatialPE-B (82.8% vs. 53.1%) but worse generation quality. *Right:* WebSSL-1B (Fan et al., 2025) likewise has higher global performance than SpatialPE-B (76.0% vs. 53.1%) but worse generation. **Spatial Self-Similarity:** The spatial metric gives the favorable ordering for generation quality, motivating the analysis in §3. All results use REPA at 100K with SiT-XL/2.

This distinction matters because it determines which quantitative evidence should guide target-encoder selection. The common framing treats ImageNet-1K validation accuracy as the primary proxy for useful global semantics (Oquab et al., 2024; Chen et al., 2021); under that view, a target representation with higher ImageNet performance should also yield better generation. For REPA, this implies that stronger ImageNet performance should identify stronger generation targets (Yu et al., 2024). Improvements in linear probing accuracy of diffusion features are therefore often interpreted as evidence that representation alignment works by transferring global semantic information. The following quote from REPA (Yu et al., 2024) captures the current understanding:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

We revisit this assumption with a comparison-first empirical study that separates global accuracy from spatial organization. Across 27 vision encoders, including recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024), we compare ImageNet-1K accuracy, spatial self-similarity metrics, and generation FID under matched REPA settings. This analysis yields 3 consistent findings about which evidence is most predictive for representation alignment.

Higher validation accuracy does not imply a better representation for generation. The encoder comparisons show clear cases where the ImageNet-accuracy ranking and the generation-FID ranking disagree. PE-Spatial-B, a spatially tuned model derived from PE-Core-G (Bolya et al., 2025), has substantially lower patch-token validation accuracy¹ than PE-Core-G (53.1% vs. 82.8%), yet produces better REPA generation quality (Figure 2). The same pattern appears when WebSSL-1B (Fan et al., 2025) is compared with PE-Spatial-B: WebSSL-1B has higher global performance (76.0% vs. 53.1%) but worse generation. In Section §2, we show that these reversals are representative of a broader mismatch between global accuracy and REPA effectiveness.

Spatial structure, not global information, determines generation performance. We therefore quantify the spatial self-similarity structure (Shechtman & Irani, 2007) of patch tokens and compare it directly against linear probing as a predictor of REPA FID (§3). Across 27 encoders and 3 diffusion model sizes (SiT-B, SiT-L, SiT-XL), all spatial structure metrics correlate much more strongly with generation FID (Pearson $|r| > 0.852$) than ImageNet-1K validation accuracy ($|r| = 0.26$). The same quantitative pattern explains several controlled observations: SAM2 obtains better REPA generation than encoders with $\sim 60\%$ higher ImageNet accuracy (Fig. 3), and even classical spatial features such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005) can deliver measurable REPA gains despite lacking the global semantic strength of modern vision encoders (§3).

Accentuating spatial features improves convergence performance. We use the correlation evidence to design two minimal changes, together requiring <4 lines of code, that make REPA transfer spatial

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

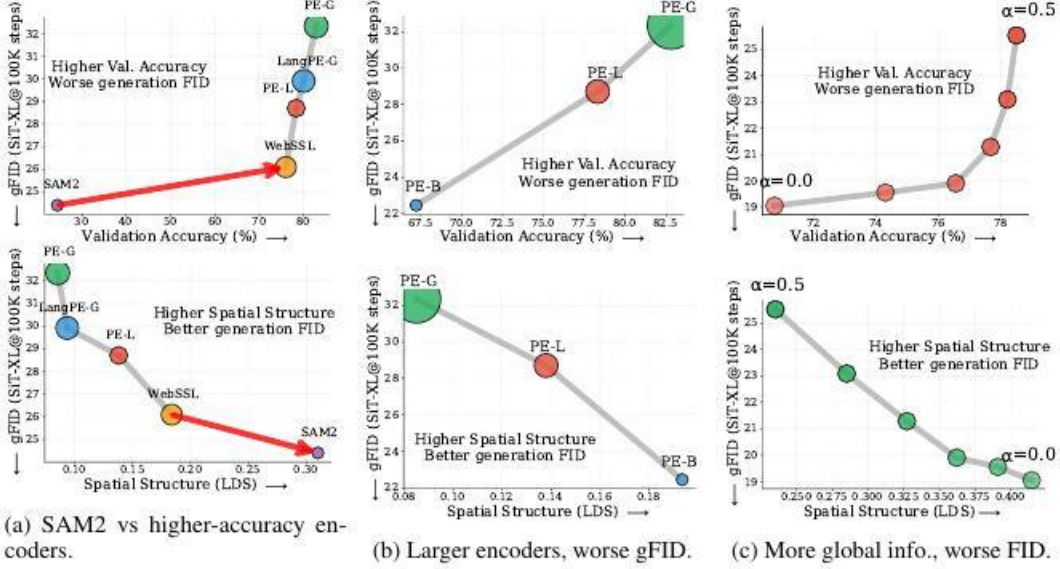


Figure 3: **Higher global information does not imply better REPA performance.** Top row: Three controlled comparisons show that global performance is poorly aligned with REPA FID. (a) SAM2-S, despite only 24.7% validation accuracy, gives better REPA generation than models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can improve validation accuracy while worsening generation. (c) Mixing CLS-token global information into patch tokens improves global performance but worsens FID. Bottom row: The corresponding spatial-structure plots provide the more consistent generation-quality signal; §3 reports the full metric analysis. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

information more directly. First, a spatial regularization layer increases spatial contrast in the target patch tokens. Second, because the standard MLP projection used to map diffusion features into the target-representation dimension reduces spatial contrast (Figure 6a), we replace it with a convolutional projection layer. The resulting method, iREPA, improves REPA convergence consistently across target encoders, model sizes, and training variants, including REPA-E (Leng et al., 2025a) and Meanflow (Geng et al., 2025) w/ REPA.

Our contributions foreground the empirical comparisons that support this reframing:

- We provide a large-scale empirical analysis showing that spatial structure, rather than global semantic accuracy, is the stronger quantitative predictor of representation-alignment effectiveness.
- We introduce a fast Spatial Structure Metric (SSM) and show that it has substantially higher correlation with downstream FID than linear probing scores across the evaluated encoders and model scales.
- We propose simple spatially focused modifications that improve REPA convergence across target representations, diffusion architectures, and training recipes, including REPA, REPA-E, Meanflow w/ REPA, and JiT w/ REPA.

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with motivating comparisons in which a target representation with higher global performance, measured by ImageNet-1K accuracy, does not translate into better REPA generation performance. These examples isolate the empirical mismatch between the commonly used global metric and the downstream FID outcome; in §3, we show that the same patterns are better explained by spatial structure in the target patch tokens.

Recent vision encoders. We compare recent encoder families that differ substantially in scale and training objective, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). The clearest reversal appears in Figure 2: PE-Core-G (1.88B) has much higher ImageNet-1K accuracy than the smaller spatially tuned PE-Spatial-B (80M) (Bolya et al., 2025), 82.8% vs. 53.1%, but gives worse REPA generation quality, FID 32.3 vs. 21.0. WebSSL-1B

(1.2B) shows the same evidence pattern: despite higher ImageNet-1K accuracy than PE-Spatial-B (76.0% vs. 53.1%), it also performs worse as the REPA target representation (FID 26.1 vs. 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To test the low-global-accuracy regime, we evaluate the SAM2-S vision encoder (46M) (Ravi et al., 2024), which has only 24.1% validation accuracy (Fig. 3a). Despite this weak global score, SAM2-S achieves better FID under REPA than encoders with far higher ImageNet-1K accuracy, including PE-Core-G at 82.8%. The comparison shows that a large global-accuracy margin alone is not sufficient evidence for better generation transfer.

Larger models within same encoder family can have similar or worse generation performance. Within an encoder family, larger variants usually improve ImageNet-1K accuracy, so they provide a controlled setting for testing whether global gains carry over to REPA. Figure 3b shows that they often do not: larger models yield similar generation performance for DINOv2 and worse generation performance for PE and Cradio. (Yu et al., 2024) report a related DINOv2 observation and attribute it to distillation from DINOv2-g; in §3, we show that the cross-family pattern is more consistently predicted by spatial structure.

Adding more global information can hurt generation. We next create a controlled intervention that increases global information in DINOv2 patch tokens by mixing in the CLS token. The operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. As α increases from 0 to 0.5, linear probing accuracy improves monotonically from 70.7% to 78.5%, but generation quality worsens: FID increases from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$ (Figure 3c). This controlled margin shows that adding global signal can move the global metric in the favorable direction while moving FID in the unfavorable direction.

Together, these comparisons show that global performance is an unreliable standalone indicator of REPA effectiveness. We next show that spatial structure provides the more favorable and consistent quantitative signal for predicting which targets improve generation, as formalized in §3.

- **Finding 1.** *Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.*

3 SPATIAL STRUCTURE MATTERS MORE

We hypothesize that spatial structure, rather than global information, is the target-representation property that best predicts REPA generation performance. To test this quantitatively, we define simple spatial self-similarity structure metrics (Shechtman & Irani, 2007) for target patch tokens and compare their correlation with gFID against the standard ImageNet-1K linear-probing baseline. Across encoders and model scales, the spatial metrics produce much larger correlations with generation quality and also explain the reversals documented in §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extract patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{(\mathbf{x}_t, \mathbf{x}_{t'})}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

We measure *spatial* self-similarity by asking how the token-similarity kernel $K_{\mathcal{X}}$ changes as lattice distance d increases. A larger score means that nearby patches are, on average, more similar to one another than distant patches, which indicates stronger spatial organization. Our default metric is a simple correlogram contrast between local and distant patch pairs (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is the expectation over patch representation $\mathcal{X}$. We use $r_{\text{near}} = r_{\text{far}} = H/2$ by default and find that the correlation conclusions are robust to these hyperparameter choices. Appendix B gives the exact definitions and evaluates alternative spatial metrics (CDS, SRSS, RMSC), which show the same favorable correlation pattern (see Fig. 4).

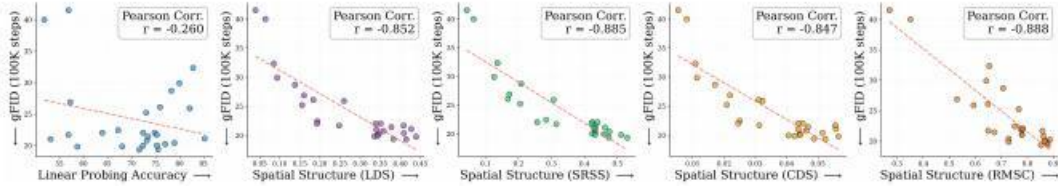


Figure 4: **Spatial structure shows higher correlation with generation quality than linear probing.** Across 27 vision encoders evaluated with SiT-XL/2 and REPA, linear probing has weak correlation with FID (Pearson $|r| = 0.260$). In contrast, every spatial structure metric is much stronger: LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$). See Fig. 10 for labeled encoder plots.

Spatial structure correlates much higher with generation performance than linear probing. We compute correlations across 27 diverse vision encoders under the same REPA evaluation setting. Figure 4 shows a large separation between the two evidence types: the standard linear-probing metric has weak correlation with generation performance (Pearson $|r| = 0.26$), whereas every spatial self-similarity metric has much higher correlation (Pearson $|r| > 0.85$).

Generalization across model scales. The correlation advantage persists when the diffusion model scale changes. In Figure 5, linear probing remains weak across SiT-B, SiT-L, and SiT-XL ($|r| < 0.306$), while the spatial structure metric maintains much stronger association with generation performance ($|r| > 0.826$). Thus, the spatial signal is not tied to a single model size.

- **Finding 2.** *Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

Spatial structure can explain previously unexplained trends. The comparisons in §2 showed that ImageNet-1K accuracy is not a reliable predictor of REPA effectiveness. Re-evaluating the same cases with spatial metrics gives a more consistent ordering.

(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows that PE-Core-G has substantially higher ImageNet-1K accuracy (82.8% vs 53.1%) but worse REPA generation quality (FID 32.3 vs 22.0). The spatial metric in Figure 2 reverses the interpretation in the useful direction: PE-Spatial-B has stronger spatial structure than PE-Core-G, matching its better FID despite lower global accuracy.

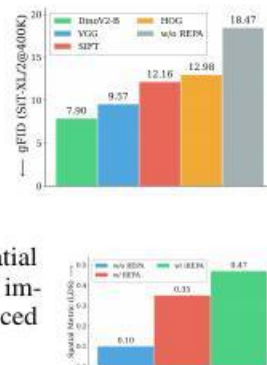
(2) *SAM2 outperforms “better” vision encoders:* §2 shows that SAM2 has much lower ImageNet-1K accuracy (24.1%) yet better generation than encoders with $\sim 60\%$ higher accuracy. Fig. 3a shows that SAM2’s stronger spatial structure accounts for this favorable FID outcome.

(3) *Larger models in same encoder family underperform:* Fig. 3b shows that larger variants within the same encoder family can improve ImageNet-1K accuracy while degrading spatial structure. The generation results follow the spatial metric rather than the global-accuracy ranking, explaining why larger PE and Cradio variants can underperform with REPA and aligning with related observations from Yu et al. (2024).

(4) *Adding global information to patch tokens via CLS token hurts generation:* In Figure 3c, increasing CLS-token mixing improves global performance but reduces spatial contrast among patch tokens. The intervention makes unrelated regions, such as foreground objects and background, more similar to each other; the resulting loss of spatial structure tracks the observed FID degradation.

If spatial structure matters more, can we use SIFT or HOG features for REPA? The answer is yes in the measured setting. Although classical spatial descriptors are not the strongest targets, SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014) still improve REPA performance. This result provides an additional check on the mechanism: representation alignment can gain from spatially organized features even without adding modern high-accuracy global semantics.

Can spatial metrics explain gains with REPA? Yes. The same spatial metrics explain both the gains from standard REPA and the additional improvements from the spatially focused training recipe, iREPA, introduced



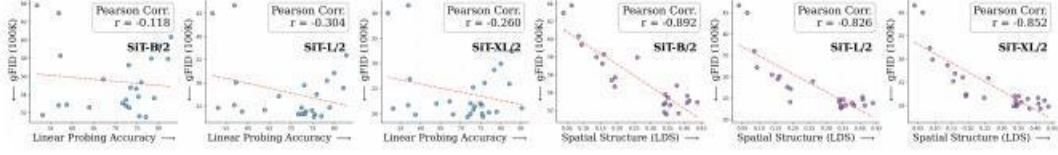


Figure 5: **Correlation analysis across model scales.** For each diffusion model scale, spatial structure (right) maintains a stronger correlation with gFID than linear probing (left), showing that the comparison advantage is not specific to one model size.

next in §4. This gives a single quantitative framing for baseline REPA benefits and for the proposed modifications.

4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

The correlation and controlled-comparison results suggest a concrete design principle: REPA should transfer spatial structure from the teacher vision encoder to the student diffusion transformer as directly as possible. We therefore introduce two small modifications to the original REPA recipe, each aimed at preserving or increasing the spatial contrast that the preceding experiments identified as predictive of generation quality.

Convolutional projection layer instead of MLP. Standard REPA uses a 3-layer MLP projection to map diffusion features into the dimensionality of the external representation. Figure 6a shows that this projection is lossy for the spatial evidence we care about: it diminishes contrast between patch-token similarities. We replace the MLP with a lightweight convolutional layer (kernel size 3, padding 1) that operates on the spatial grid, so the projection retains a local inductive bias while matching the same feature dimensions.

Spatial normalization layer. Similar to Siméoni et al. (2025), we find that pretrained-encoder patch tokens contain a substantial global component. The evidence is quantitative and visual: the mean of patch tokens obtains high linear-probing scores (Figure 14), and Figure 6b shows that mixing this global mean information into patch tokens improves global performance while reducing spatial contrast. In practice, tokens from unrelated regions, such as foreground objects and background, become more similar.

Given the results in §3, we intentionally trade off some global mean information to increase spatial contrast among patch tokens. This should strengthen the pairwise spatial signal that best predicts REPA performance. We implement the change with a simple spatial normalization layer (Ulyanov et al., 2016) applied to the target-representation patch tokens:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ represents the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ for numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

4.1 EXPERIMENTS

We evaluate whether the spatially focused iREPA modifications improve the same metrics and settings used for REPA, with experiments on Imagenet 256×256 . The comparisons are organized around the baseline method, target encoder, model scale, training recipe, and evaluation condition:

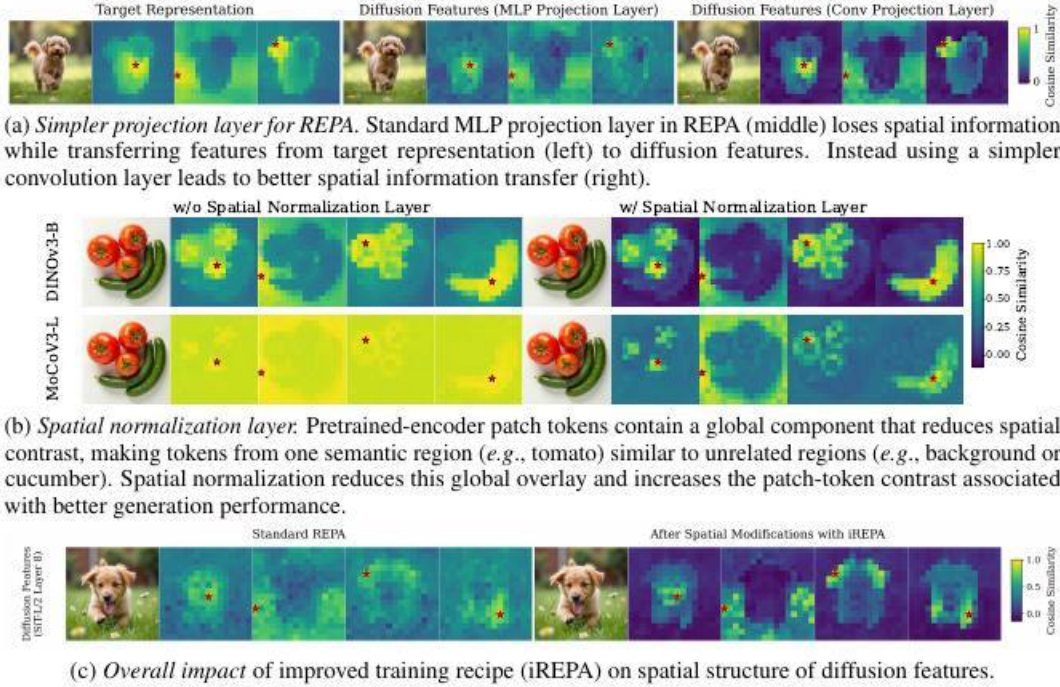


Figure 6: Motivating the two iREPA modifications with spatial evidence from the REPA baseline. The convolutional projection preserves spatial structure more directly than the MLP projection, while spatial normalization increases patch-token contrast. All results use SiTL/2 with REPA at 100K.

- Does iREPA improve convergence speed over REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)
- Do the gains persist across model size, encoder size, and alignment depth? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Do the same spatial modifications improve newer representation-alignment and diffusion settings, including REPA-E (Leng et al., 2025a), MeanFlow w/ REPA (Geng et al., 2025), and JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We first compare REPA and iREPA convergence for the same target encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and model sizes (SiT-XL/2 and SiT-B/2). Figure 7 shows a consistent empirical pattern: when the baseline REPA setting is held fixed, iREPA reaches better FID earlier across both target-representation and model-size variations.

Target representation. We next evaluate the encoder dimension directly in Fig. 8 and Table 4. Across the tested vision encoders, iREPA improves generation quality relative to the corresponding REPA baseline. Appendix C, E reports the full set of comparisons across all 27 encoders.

Enc. Size	IS↑	FID↓	sFID↓	Pr.↑	Rc.↑
PE-B (90M)	59.6	22.5	6.23	0.63	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60
+iREPA	76.3	17.6	6.36	0.65	0.61
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59
+iREPA	70.8	19.5	6.35	0.64	0.61

(a) Vision Encoder size

Model Size	IS↑	FID↓	sFID↓	Pr.↑	Rc.↑
SiT-B	27.5	49.50	7.00	0.46	0.59
+iREPA	34.1	43.37	7.87	0.50	0.60
SiT-L	55.7	24.10	6.25	0.62	0.60
+iREPA	66.9	20.28	6.14	0.63	0.62
SiT-XL	70.3	19.06	5.83	0.65	0.61
+iREPA	77.9	16.96	6.26	0.66	0.61

(b) Model size

Aln. Depth	IS↑	FID↓	sFID↓	Pr.↑	Rc.↑
Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	41.7	36.0	6.61	0.52	0.62
Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	42.4	35.3	6.90	0.53	0.62
Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	38.3	38.9	7.19	0.51	0.62

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA is compared against the matching REPA baseline across encoder size, diffusion model size, and alignment depth. Unless otherwise specified, results use DINOv2 as encoder, SiT-XL/2, 100K steps, and vanilla REPA as the baseline.

Encoder size. Table 1a compares iREPA against REPA across PE encoder sizes: PE-B (90M), PE-L (320M), and PE-G (1.88B). The gains are consistent for all three target sizes, and the percentage FID improvement increases with encoder scale: 22.2% for PE-B, 38.8% for PE-L, and 39.6% for PE-G.

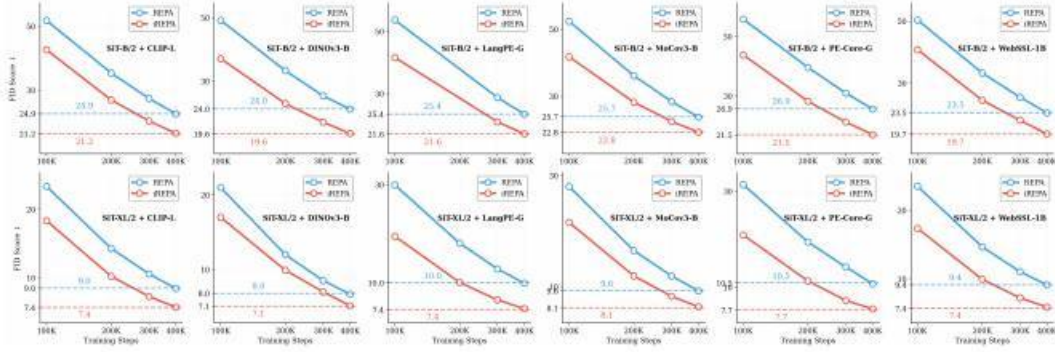


Figure 7: **Accentuating spatial features improves convergence speed.** FID curves compare baseline REPA with iREPA across target encoders and model sizes, showing faster convergence when spatial transfer is emphasized.

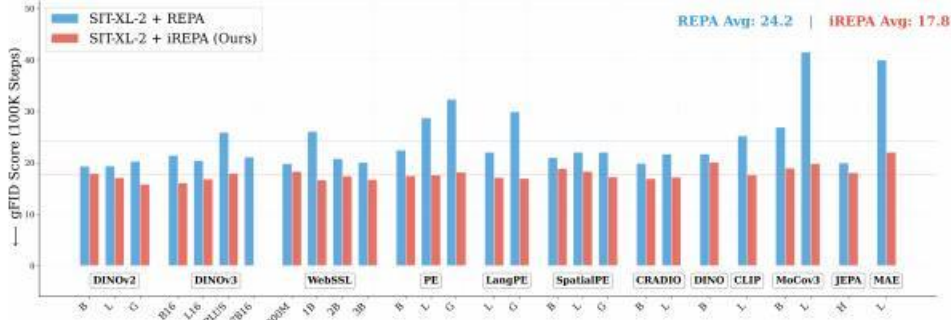


Figure 8: **Variation in target representation.** Across all 27 vision encoders, iREPA improves the corresponding REPA baseline by accentuating spatial transfer from the target representation to diffusion features. Appendix C provides the full set of comparisons.

Scalability. Table 1b holds the spatially focused method fixed and varies the diffusion model scale. iREPA improves over REPA at each scale, with larger percentage gains for larger models, indicating that the spatial modifications remain effective as the student diffusion transformer scales.

Encoder depth. Table 1c tests alignment depth with SiT-B/2 at 100K iterations using DINOv3-B as the target representation. Across the evaluated depths, iREPA improves the baseline REPA comparison, showing that the effect is not limited to a single alignment layer.

Ablation on different components. Table 2 separates the two iREPA components. Spatial normalization and the convolutional projection layer each improve generation quality over baseline REPA, and the best result is obtained when both components are used together, matching the intended design of increasing spatial contrast while preserving local structure.

Training recipe. We also test whether the same spatial evidence transfers beyond vanilla REPA. Table 3 applies the iREPA modifications on top of REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025); in both recipes, the spatial modifications yield convergence gains under the reported 100K-iteration setting.

Classifier-free guidance. Table 4 compares REPA and iREPA with and without CFG across encoders and training budgets. The pattern is consistent across evaluation conditions: iREPA improves convergence without CFG and preserves the advantage when classifier-free guidance is applied.

Pixel-space diffusion (JiT). Finally, we evaluate whether the spatially focused modification carries over to pixel-space diffusion. Figure 9 compares REPA and iREPA with JiT-B (Li & He, 2025b) across encoders such as DINOv2, DINOv3, and PE. iREPA again improves convergence with JiT

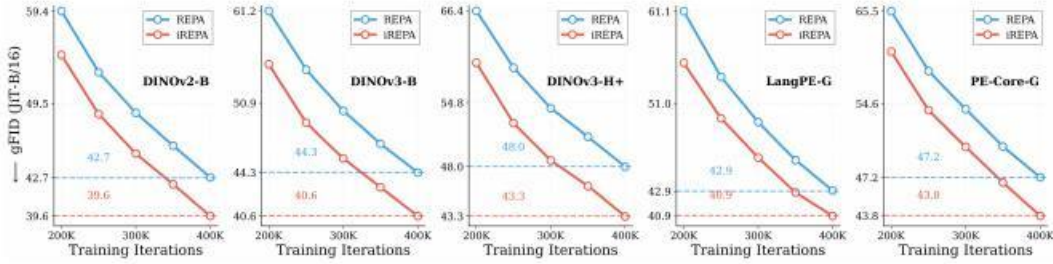


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID curves compare REPA and iREPA with pixel-space JiT (Li & He, 2025b) across vision encoders. The same spatial modification consistently improves convergence in this pixel-space setting. All results use JiTB/16 (Li & He, 2025b), batch size 256, and no classifier-free guidance; Table 10 provides additional results.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on iREPA components.** Across encoders, spatial normalization and the convolutional projection layer (§4) each improve the REPA baseline, and using both gives the best convergence result. All results are reported at 100K steps with SiT-XL/2.

(Li & He, 2025b), indicating that the benefit of accentuating spatial transfer is not restricted to latent-space diffusion.

- **Finding 3.** *Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).*

5 RELATED WORK

We focus here on the related work most directly connected to the empirical comparisons in this paper, with additional discussion in Appendix H.

Representation alignment for generation. Recent methods use external representations to improve diffusion-model training (Pernias et al., 2023; Fuest et al., 2024). In particular, REPA-style approaches (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) report large gains by aligning diffusion features with clean-image features from pretrained vision encoders. The same idea has also been extended to video and 3D generation (Zhang et al., 2025; Wu et al., 2025), and to unified models (Ma et al., 2025a). Our work keeps the empirical success of alignment fixed and asks which measured property of the target encoder explains the gains: global semantic accuracy or spatial structure. We show that the spatial metric provides the clearer quantitative account and use that evidence to improve the training recipe.

Spatial vs global information tradeoff in pretrained vision encoders. Recent studies examine how pretrained vision encoders trade global semantics against spatial information (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued self-supervised training can increase similarity between the global CLS token and patch tokens, with negative consequences for dense spatial tasks. (Bolya et al., 2025; Heinrich et al., 2024) train spatially tuned models for dense prediction. Our results extend this evidence to generative training: across the evaluated REPA comparisons, spatial structure is a stronger predictor of generation quality than global ImageNet-1K accuracy. This suggests that external representations for generation should be selected and trained using metrics that foreground spatial organization, not global classification performance alone.

Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** The same spatial modifications are applied to REPA-E (a) and Meanflow with REPA (b), and each row compares against the corresponding recipe baseline. For Meanflow, results are reported at 1 and 4 NFEs with CFG value 2.0. All results are reported at 100K training iterations.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **REPA vs iREPA across encoders with and without classifier-free guidance.** SiT-XL/2 is the base model, and each +iREPA row is compared to the corresponding REPA encoder, training-step, and CFG condition. See Appendix E for SiT-B/2 & SiT-L/2 results across all encoders.

6 CONCLUSION

We studied which quantitative property makes a target representation effective for representation alignment: global information or spatial structure. Across 27 encoders, multiple model scales, and controlled interventions, the empirical evidence consistently favors spatial structure. ImageNet-1K accuracy and linear probing can move in the opposite direction from REPA generation quality, whereas spatial self-similarity metrics show much stronger correlation with FID and explain the main comparison patterns. Building on this evidence, we introduced iREPA, a minimal modification that uses spatial normalization and a convolutional projection layer to accentuate spatial transfer from the target representation to diffusion features. iREPA improves REPA convergence across diverse encoders, model sizes, classifier-free-guidance settings, and training recipes. We hope these results motivate future work to evaluate and design alignment targets using spatial evidence that is directly tied to practical generation performance.

REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

REFERENCES

- Kristian Abstreiter, Sarthak Mittal, Stefan Bauer, Bernhard Schölkopf, and Arash Mehrjou. Diffusion-based representation learning. In *ICML 2021 Workshop on Unsupervised Reinforcement Learning*, 2021.
- Stability AI. Improved autoencoders ... <https://huggingface.co/stabilityai/sd-vae-ft-mse>, n.d. Accessed: April 11, 2025.
- Anthropic. Claude code. <https://claude.ai/code>, 2025.
- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15619–15629, 2023.
- Fan Bao, Shen Nie, Kaiwen Xue, Chongxuan Li, Shi Pu, Yaole Wang, Gang Yue, Yue Cao, Hang Su, and Jun Zhu. All are worth words: A vit backbone for diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 22669–22679, 2023.
- Herbert Bay, Tinne Tuytelaars, and Luc Van Gool. Surf: Speeded up robust features. In *Computer Vision—ECCV 2006: 9th European Conference on Computer Vision, Graz, Austria, May 7–13, 2006. Proceedings, Part I* 9, pp. 404–417. Springer, 2006.
- Daniel Bolya, Po-Yao Huang, Peize Sun, Jang Hyun Cho, Andrea Madotto, Chen Wei, Tengyu Ma, Jiale Zhi, Jathushan Rajasegaran, Hanoona Rasheed, Junke Wang, Marco Monteiro, Hu Xu, Shiyu Dong, Nikhila Ravi, Daniel Li, Piotr Dollár, and Christoph Feichtenhofer. Perception encoder: The best visual embeddings are not at the output of the network. *arXiv:2504.13181*, 2025.
- Shoufa Chen, Mengmeng Xu, Jiawei Ren, Yuren Cong, Sen He, Yanping Xie, Nicu Sebe, and Wai Lam. Gentron: Diffusion transformers for image and video generation. *arXiv preprint arXiv:2312.04557*, 2023.
- Shoufa Chen, Chongjian Ge, Shilong Zhang, Peize Sun, and Ping Luo. Pixelflow: Pixel-space generative models with flow. *arXiv preprint arXiv:2504.07963*, 2025.
- Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 9640–9649, 2021.
- Xinlei Chen, Zhuang Liu, Saining Xie, and Kaiming He. Deconstructing denoising diffusion models for self-supervised learning. *arXiv preprint arXiv:2401.14404*, 2024.
- Jang Hyun Cho, Andrea Madotto, Effrosyni Mavroudi, Triantafyllos Afouras, Tushar Nagarajan, Muhammad Maaz, Yale Song, Tengyu Ma, Shuming Hu, Hanoona Rasheed, Peize Sun, Po-Yao Huang, Daniel Bolya, Suyog Jain, Miguel Martin, Huiyu Wang, Nikhila Ravi, Shashank Jain, Temmy Stark, Shane Moon, Babak Damavandi, Vivian Lee, Andrew Westbury, Salman Khan, Philipp Krähenbühl, Piotr Dollár, Lorenzo Torresani, Kristen Grauman, and Christoph Feichtenhofer. Perceptionlm: Open-access data and models for detailed visual understanding. *arXiv:2504.13180*, 2025.
- N. Dalal and B. Triggs. Histograms of oriented gradients for human detection. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR’05)*, volume 1, pp. 886–893 vol. 1, 2005. doi: 10.1109/CVPR.2005.177.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pp. 248–255. Ieee, 2009.

- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first International Conference on Machine Learning*, 2024.
- David Fan, Shengbang Tong, Jiachen Zhu, Koustuv Sinha, Zhuang Liu, Xinlei Chen, Michael Rabbat, Nicolas Ballas, Yann LeCun, Amir Bar, et al. Scaling language-free visual representation learning. *arXiv preprint arXiv:2504.01017*, 2025.
- Michael Fuest, Pingchuan Ma, Ming Gui, Johannes Schusterbauer, Vincent Tao Hu, and Bjorn Ommer. Diffusion models and representation learning: A survey. *arXiv preprint arXiv:2407.00783*, 2024.
- Zhengyang Geng, Mingyang Deng, Xingjian Bai, J Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling. *arXiv preprint arXiv:2505.13447*, 2025.
- Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks. *Communications of the ACM*, 63(11):139–144, 2020.
- Greg Heinrich, Mike Ranzinger, Hongxu, Yin, Yao Lu, Jan Kautz, Andrew Tao, Bryan Catanzaro, and Pavlo Molchanov. Radiov2.5: Improved baselines for agglomerative vision foundation models, 2024. URL <https://arxiv.org/abs/2412.07679>.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- Jing Huang, S.R. Kumar, M. Mitra, Wei-Jing Zhu, and R. Zabih. Image indexing using color correlograms. In *Proceedings of IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, pp. 762–768, 1997. doi: 10.1109/CVPR.1997.609412.
- Drew A Hudson, Daniel Zoran, Mateusz Malinowski, Andrew K Lampinen, Andrew Jaegle, James L McClelland, Loic Matthey, Felix Hill, and Alexander Lerchner. Soda: Bottleneck diffusion models for representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 19461–19471, 2024.
- Sadeep Jayasumana, Srikumar Ramalingam, Andreas Veit, Daniel Glasner, Ayan Chakrabarti, and Sanjiv Kumar. Rethinking fid: Towards a better evaluation metric for image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9307–9315, June 2024.
- Dengyang Jiang, Mengmeng Wang, Liuzhuozheng Li, Lei Zhang, Haoyu Wang, Wei Wei, Guang Dai, Yanning Zhang, and Jingdong Wang. No other representation component is needed: Diffusion transformers can provide representation guidance by themselves. *arXiv preprint arXiv:2505.02831*, 2025.
- Jiatao Jing, Gu Wu, Yuanpeng Zhou, Yuxin Xu, Zixun Sun, Xinhao Ni, Yazhou Wang, Chunmeng Li, Hao Tang, Ruihua Chen, et al. Dart: Denoising autoregressive transformers for scalable text-to-image generation. *arXiv preprint arXiv:2410.08159*, 2024.
- Minguk Kang, Richard Zhang, Connelly Barnes Nga Tran, Saurabh Kar, and Jun-Yan Zhu. Distilling diffusion models into conditional gans. *arXiv preprint arXiv:2405.05967*, 2024.
- Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- Theodoros Kouzelis, Efstathios Karypidis, Ioannis Kakogeorgiou, Spyros Gidaris, and Nikos Komodakis. Boosting generative image modeling via joint image-feature synthesis. *arXiv preprint arXiv:2504.16064*, 2025.

- Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. *Advances in neural information processing systems*, 32, 2019.
- Tuomas Kynkäänniemi, Tero Karras, Miika Aittala, Timo Aila, and Jaakko Lehtinen. The role of imagenet classes in fréchet inception distance. In *International Conference on Learning Representations*, 2023.
- Xingjian Leng, Jaskirat Singh, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. Repa-e: Unlocking vae for end-to-end tuning with latent diffusion transformers. *arXiv preprint arXiv:2504.10483*, 2025a.
- Xingjian Leng, Jaskirat Singh, Ryan Murdock, Ethan Smith, Rebecca Li, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. Family of End-to-End Tuned VAEs for Supercharging T2I Diffusion Transformers. <https://end2end-diffusion.github.io/repa-e-t2i/>, 2025b.
- Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv preprint arXiv:2511.13720*, 2025a.
- Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv preprint arXiv:2511.13720*, 2025b.
- I Loshchilov. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- David G Lowe. Object recognition from local scale-invariant features. In *Proceedings of the seventh IEEE international conference on computer vision*, volume 2, pp. 1150–1157. Ieee, 1999.
- Nanye Ma, Mark Goldstein, Michael S Albergo, Nicholas M Boffi, Eric Vanden-Eijnden, and Saining Xie. Sit: Exploring flow and diffusion-based generative models with scalable interpolant transformers. In *European Conference on Computer Vision*, pp. 23–40. Springer, 2024.
- Yiyang Ma, Xingchao Liu, Xiaokang Chen, Wen Liu, Chengyue Wu, Zhiyu Wu, Zizheng Pan, Zhenda Xie, Haowei Zhang, Xingkai Yu, et al. Janusflow: Harmonizing autoregression and rectified flow for unified multimodal understanding and generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 7739–7751, 2025a.
- Zehong Ma, Longhui Wei, Shuai Wang, Shiliang Zhang, and Qi Tian. Deco: Frequency-decoupled pixel diffusion for end-to-end image generation. *arXiv preprint arXiv:2511.19365*, 2025b.
- Charlie Nash, Jacob Menick, Sander Dieleman, and Peter Battaglia. Generating images with sparse representations. In *International Conference on Machine Learning*, pp. 7958–7968. PMLR, 2021.
- OpenAI. Gpt-5. <https://openai.com>, 2025.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research Journal*, pp. 1–31, 2024.
- Sayak Paul. Cmmd-pytorch: Pytorch implementation of cmmd metric. <https://github.com/sayakpaul/cmmd-pytorch>, 2024. Accessed: 2025.
- William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4195–4205, 2023.
- Pablo Pernias, Dominic Rampas, Mats Leon Richter, Christopher Pal, and Marc Aubreville. Würstchen: An efficient architecture for large-scale text-to-image diffusion models. In *The Twelfth International Conference on Learning Representations*, 2023.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PMLR, 2021.

- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. URL <https://arxiv.org/abs/2408.00714>.
- Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. Orb: An efficient alternative to sift or surf. In *2011 International conference on computer vision*, pp. 2564–2571. Ieee, 2011.
- Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training gans. *Advances in neural information processing systems*, 29, 2016.
- Axel Sauer, Katja Schwarz, and Andreas Geiger. Projected gans converge faster. In *Advances in Neural Information Processing Systems*, volume 34, pp. 17480–17492, 2021.
- Axel Sauer, Katja Schwarz, and Andreas Geiger. Stylegan-xl: Scaling stylegan to large diverse datasets. In *ACM SIGGRAPH 2022 conference proceedings*, pp. 1–10, 2022.
- Axel Sauer, Tero Karras, Samuli Laine, Andreas Geiger, and Timo Aila. Stylegan-t: Unlocking the power of gans for fast large-scale text-to-image synthesis. *arXiv preprint arXiv:2301.09515*, 2023.
- Axel Sauer, Dominik Lorenz, Andreas Blattmann, and Robin Rombach. Adversarial diffusion distillation. In *European Conference on Computer Vision*, pp. 218–234. Springer, 2024.
- Eli Shechtman and Michal Irani. Matching local self-similarities across images and videos. In *2007 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–8, 2007. doi: 10.1109/CVPR.2007.383198.
- Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. URL <https://arxiv.org/abs/2508.10104>.
- Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.
- George Stein, Jesse C. Cresswell, Rasa Hosseinzadeh, Yi Sui, Brendan Leigh Ross, Valentin Villecroze, Zhaoyan Liu, Anthony L. Caterini, J. Eric T. Taylor, and Gabriel Loaiza-Ganem. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models. In *Advances in Neural Information Processing Systems*, 2023.
- Shitao Tang, Fuyang Chen, Aviral Joshi Bose, Kun Zhang, Shengjie Bi, Zhenyu Wang, Jiacheng Feng, Haozhe Cai, Alexander Liu, Antonio Torralba, et al. Ardit: Autoregressive diffusion transformers. *arXiv preprint arXiv:2406.00843*, 2024.
- Dmitry Ulyanov, Andrea Vedaldi, and Victor Lempitsky. Instance normalization: The missing ingredient for fast stylization. *arXiv preprint arXiv:1607.08022*, 2016.
- Runqian Wang and Kaiming He. Diffuse and disperse: Image generation with representation regularization. *arXiv preprint arXiv:2506.09027*, 2025.
- Shuai Wang, Ziteng Gao, Chenhui Zhu, Weilin Huang, and Limin Wang. Pixnerd: Pixel neural field diffusion. *arXiv preprint arXiv:2507.23268*, 2025a.
- Shuai Wang, Zhi Tian, Weilin Huang, and Limin Wang. Ddt: Decoupled diffusion transformer. *arXiv preprint arXiv:2504.05741*, 2025b.
- Ziqiao Wang, Wangbo Zhao, Yuhao Zhou, Zekai Li, Zhiyuan Liang, Mingjia Shi, Xuanlei Zhao, Pengfei Zhou, Kaipeng Zhang, Zhangyang Wang, Kai Wang, and Yang You. Repa works until it doesn't: Early-stopped, holistic alignment supercharges diffusion training. *arXiv preprint arXiv:2505.16792*, 2025c.

-
- Haoyu Wu, Diankun Wu, Tianyu He, Junliang Guo, Yang Ye, Yueqi Duan, and Jiang Bian. Geometry forcing: Marrying video diffusion and 3d representation for consistent world modeling. *arXiv preprint arXiv:2507.07982*, 2025.
- Mengping Yang, Ceyuan Yang, Yichi Zhang, Qingyan Bai, Yujun Shen, and Bo Dai. Revisiting the evaluation of image synthesis with gans. In *Advances in Neural Information Processing Systems*, 2023.
- Jingfeng Yao and Xinggang Wang. Reconstruction vs. generation: Taming optimization dilemma in latent diffusion models. *arXiv preprint arXiv:2501.01423*, 2025.
- Sihyun Yu, Sangkyung Kwak, Huiwon Jang, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, and Saining Xie. Representation alignment for generation: Training diffusion transformers is easier than you think. *arXiv preprint arXiv:2410.06940*, 2024.
- Xiangdong Zhang, Jiaqi Liao, Shaofeng Zhang, Fanqing Meng, Xiangpeng Wan, Junchi Yan, and Yu Cheng. Videorepa: Learning physics for video generation through relational alignment with foundation models. *arXiv preprint arXiv:2505.23656*, 2025.

A DETAILED CORRELATION ANALYSIS

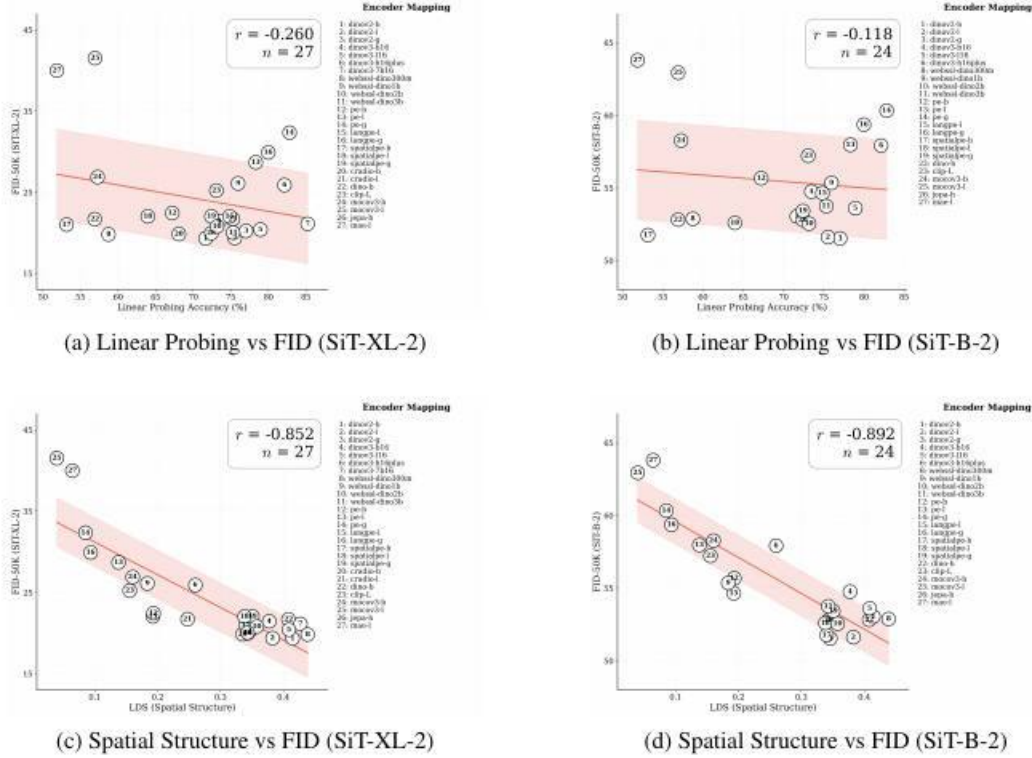


Figure 10: **Detailed correlation analysis with encoder labels.** For both SiT-XL-2 and SiT-B-2, the top row compares linear probing accuracy with FID, while the bottom row compares LDS spatial structure with FID. Each point is labeled by encoder number (see legend), making the cross-encoder comparison explicit. Across both model scales, the LDS panels show the stronger and more favorable association with generation quality.

B SPATIAL SELF-SIMILARITY METRICS

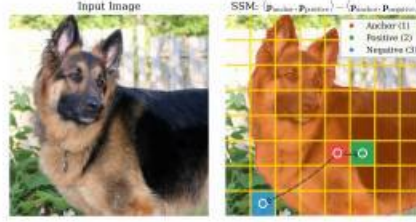


Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** For each image, SAM2 provides a mask used to form anchor-positive-negative triplets. SRSS measures the cosine-similarity margin between anchor-positive pairs inside the mask and anchor-negative pairs outside the mask. Larger values therefore mean that same-region patches are more similar than unrelated-region patches, giving a direct quantitative score for spatial structure preservation.

Setup. For each image $\mathcal{I}$ and vision encoder $\mathcal{E}$, the comparison unit is the patch-token set $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, where $T = H \times W$ tokens occupy an $H \times W$ lattice with indices $t \in [T]$. We use the Manhattan distance $d : [T] \times [T] \rightarrow \mathbb{N}$ to make spatial separation explicit, and measure token similarity with the cosine self-similarity kernel

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007). This setup lets every metric below compare similarity among nearby patches against similarity among spatially distant or semantically unrelated patches.

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ is the feature table being scored, with $T = H \times W$ patches indexed by $[T]$ on the spatial grid. The distance function $d : [T] \times [T] \rightarrow \mathbb{N}$ records Manhattan separation on that grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ gives the pairwise token similarity used for every spatial comparison (Shechtman & Irani, 2007). In all reported metrics, this is the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ converts the similarity matrix K and lattice distances d into a single spatial score. Larger scores consistently indicate stronger spatial organization, meaning nearby or same-region patches are more similar than far-away or unrelated patches.

We next define several metrics that quantify the same comparison target: how strongly a representation preserves spatial self-similarity among patch tokens.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are the hyperparameters. By default we use $r_{\text{near}} = r_{\text{far}} = H/2$, and the correlation evidence remains robust to their exact choices. Larger values indicate the practically important pattern that nearby patches are, on average, more similar than distant patches.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values mean that similarity decays faster with distance, so the metric assigns a higher score to representations with stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, SAM2 (Ravi et al., 2024) provides a binary object mask $\mathbf{M} \in \mathbb{R}^{H \times W}$, which we downsample to the token grid. The metric then forms

anchor, positive, and negative points: anchor and positive points come from within the mask and should be semantically related, while the negative point comes from outside the mask and should be less related. A representation with better spatial structure should therefore have a positive cosine-similarity margin between anchor-positive and anchor-negative pairs. We define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

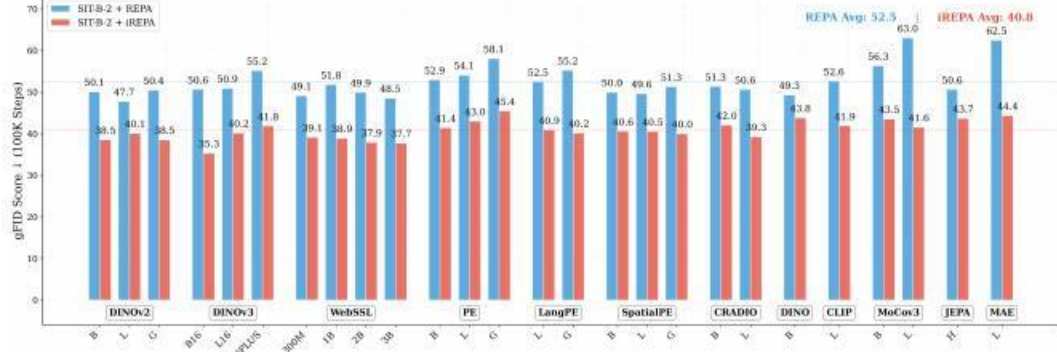
where positive points are within Manhattan distance $d \leq r_{\text{near}}$ from the anchor inside the mask, and negative points are at distance $d \geq r_{\text{far}}$ outside the mask. Larger margins indicate that semantic-region structure is more clearly preserved in the patch-token geometry.

RMS Spatial Contrast (RMSC). Finally, we use a direct dispersion score to measure how much contrast remains among patch-token representations after normalization. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

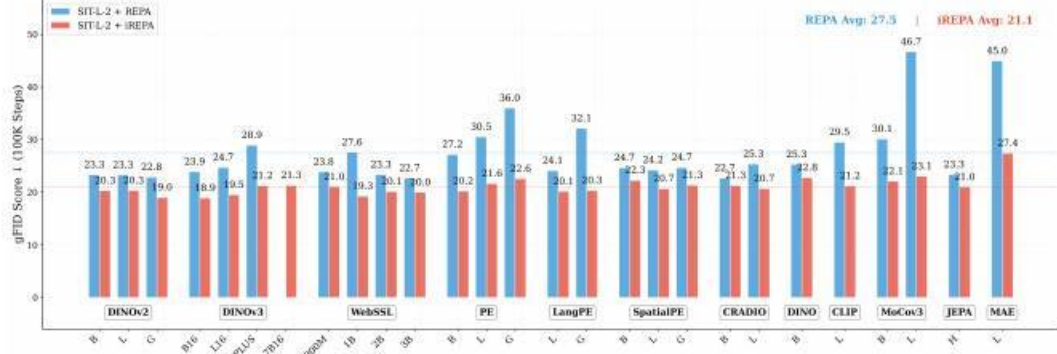
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean normalized feature across spatial locations. Higher RMSC values mean greater spatial diversity in the token representation, which favors preserved spatial structure; lower values mean the feature distribution is more uniform, consistent with the loss of spatial structure that can occur when global information diminishes patch-level contrast (Siméoni et al., 2025).

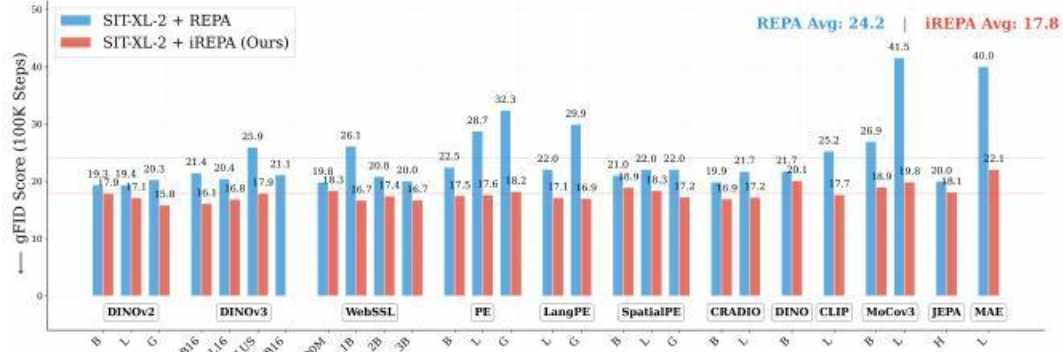
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



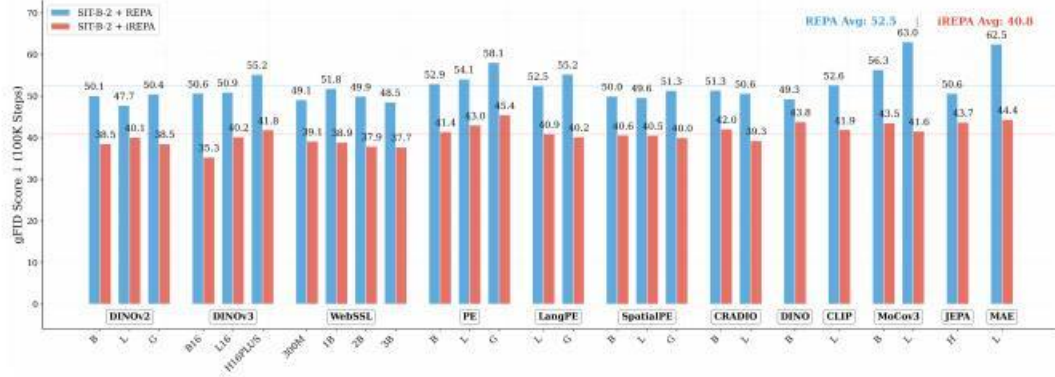
(b) SiT-L-2 (458M parameters)



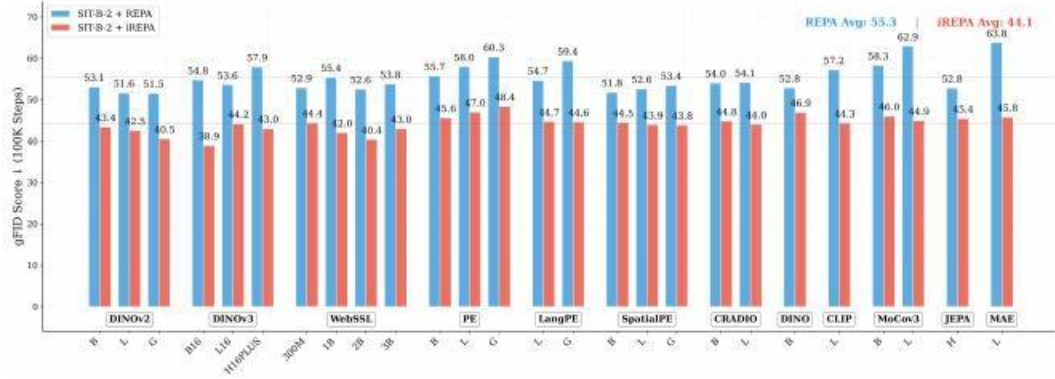
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Each panel compares iREPA against the matched REPA baseline across target encoders for SiT-B-2, SiT-L-2, and SiT-XL-2. The direction of the FID change is consistent across encoder choices and diffusion-model scales, showing that the spatially focused modification improves generation quality under the same evaluation condition.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** The figure compares SiT-B-2 with 6 and 8 encoder layers across the same vision-encoder choices. In both depth settings, iREPA improves the REPA baseline in FID, with the 8-layer configuration giving slightly better performance. The comparison shows that the spatial gains persist across alignment-depth choices.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality across all encoders, with particularly strong gains in FID and IS metrics. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

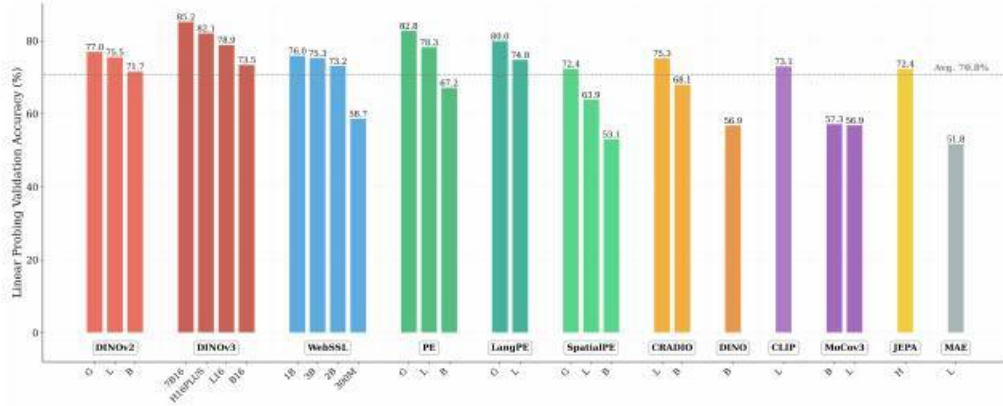


Figure 14: **Global information in mean patch tokens.** The mean of patch tokens, like the [CLS] token, carries substantial global semantic information in these encoders. This improves the global-performance metric but reduces contrast among individual patch tokens, which is the spatial signal used in our REPA comparisons. Removing part of this mean-token global component improves spatial structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** REPA and iREPA are compared with CFG scale 2.0 across vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps; iREPA improves the corresponding IS and FID comparisons across encoders.

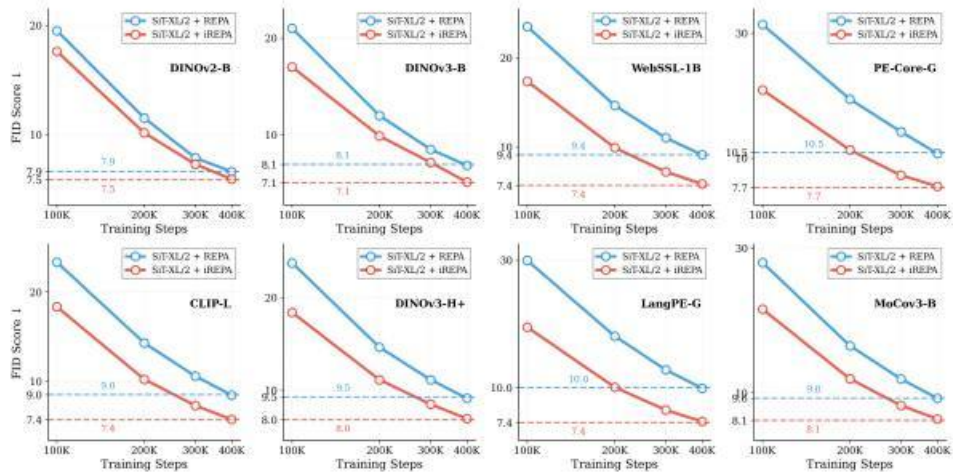


Figure 15: **Accentuating spatial features improves convergence speed.** FID convergence for SiT-XL/2 compares the REPA baseline with iREPA under the same target-encoder settings, showing faster convergence after the spatial modifications.

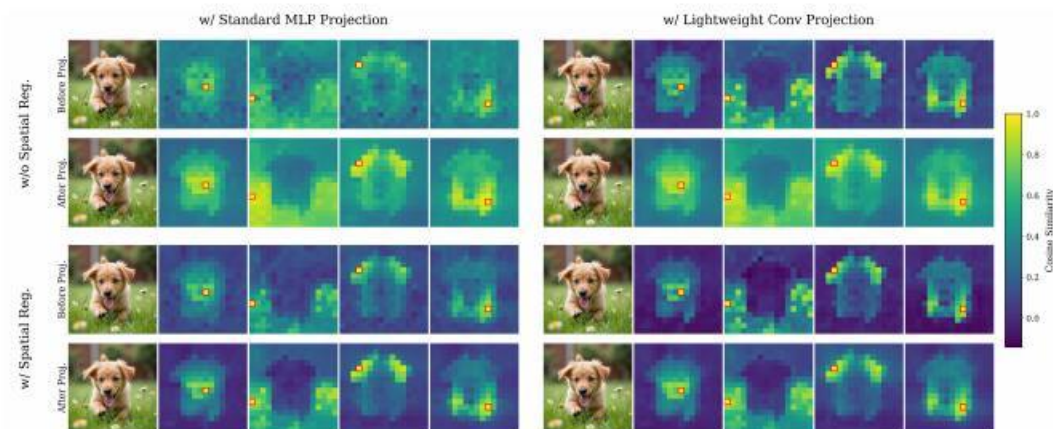


Figure 16: Visualizing the two iREPA changes that improve spatial feature transfer. First, the standard MLP projection layer (top-left) loses spatial information when mapping diffusion features to the pretrained-encoder space. A simpler convolution layer preserves the spatial transfer more clearly under the same visualization condition (top-right). Second, vision-encoder features often have limited spatial contrast (§2), so tokens from one semantic region (*e.g.*, dog) can remain highly similar to unrelated tokens (*e.g.*, background). The spatial regularization layer increases this contrast, and Table 2 shows that combining it with the convolutional projection gives the best reported generation result.

G IMPLEMENTATION DETAILS

Training setup. Unless otherwise specified, each comparison follows the REPA (Yu et al., 2024) setup so that the reported margins isolate the effect of the spatial modifications. All training uses the ImageNet (Deng et al., 2009) training split with ADM (Dhariwal & Nichol, 2021) preprocessing: each image is center-cropped and resized to 256×256 . We use the `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) for diffusion training and inference. For the iREPA components, the spatial normalization layer uses $\gamma \in [0.6, 0.8]$, and the projection layer is a convolution with kernel size 3 and padding 1. Optimization uses AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with constant learning rate 1×10^{-4} and global batch size 256. During training, `bfloat16` mixed precision, `torch.compile`, gradient clipping, and exponential moving average (EMA) keep the comparison efficient and stable.

For REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a), we use the official open-source implementations. For MeanFlow (Geng et al., 2025), we use the implementation, including REPA, provided by the original authors and change only the two iREPA components described in §4.

Evaluation. For image-generation evaluation, we strictly follow the ADM setup (Dhariwal & Nichol, 2021) and report the same metrics for each baseline/iREPA comparison: Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.), and recall (Rec.) (Kynkäänniemi et al., 2019), all measured on 50K generated images. Sampling follows REPA (Yu et al., 2024) with the SDE Euler-Maruyama sampler and 250 steps; for JiT (Li & He, 2025a), we use the official 50-step inference setting.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been fundamental to computer vision. Classical methods like SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) providing robust local descriptors for tasks ranging from object detection to image matching. While these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent work in generative modeling has primarily focused on representations from modern self-supervised methods that demonstrate strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings suggest a different perspective: since spatial structure preservation is critical for generation quality, even classical spatial features could potentially improve diffusion training when properly aligned. This highlights the potential of leveraging the full spectrum of spatial feature extractors, from traditional handcrafted descriptors to modern learned representations, provided that they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models in several capacities: as discriminators to accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and more recently as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have not only improved convergence speed but also enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation leverages pretrained encoders to guide student networks toward higher-fidelity samples, showing clear improvements in perceptual quality. In particular, REPA (Yu et al., 2024) aligns diffusion features with external encoders, demonstrating that representation alignment can improve both generation convergence and quality. Building on this direction, we focus not only on alignment but specifically on spatial structure perseverance rather than their discriminative capabilities.

Denosing transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, with various formulations including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent architectures like GenTron (Chen et al., 2023) scale transformers to over 3B parameters for text-to-image synthesis. U-ViT (Bao et al., 2023) demonstrates that pure transformer backbones without U-Net inductive biases can achieve competitive performance. ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that combine denoising with sequential generation, enabling flexible

trade-offs between quality and speed. Despite these architectural advances, training these models from scratch remains computationally expensive, often requiring millions of iterations to achieve good generation quality. While representation alignment methods like REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate convergence, the mechanisms behind this improvement remained unclear. Our analysis reveals that the key benefit comes from preserving spatial structure rather than semantic alignment, explaining why certain encoders provide stronger acceleration than others and guiding the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denosing as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. These works establish denoising as a fundamental self-supervised task that naturally encourages learning of robust features. Our findings complement this view by showing that when diffusion models are aligned with strong spatial representations from self-supervised encoders, both the generative and discriminative capabilities improve, suggesting that spatial structure preservation is a key factor in this synergy.

Scaling self-supervised vision encoders. Recent years have seen remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B parameter ViT on 1.7 billion images without labels by aligning representations from different augmentations. WebSSL (Fan et al., 2025) demonstrates that visual models trained on more than 2 billion images can match language-supervised models like CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) combines multiple teacher models through distillation to create versatile encoders that excel across diverse visual domains. I-JEPA (Assran et al., 2023) explores predictive architectures that learn by predicting masked regions in representation space, instead of reconstructing pixels directly. SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations for dense prediction tasks. While these models are typically evaluated on global tasks like image classification, our work reveals a crucial insight: strong performance on discriminative benchmarks does not necessarily translate to better generation quality. Instead, we find that encoders preserving spatial structure, regardless of their classification accuracy—provide the most benefit for diffusion training. This suggests that the evaluation metrics for self-supervised encoders should be reconsidered when targeting generative applications, with spatial coherence being as important as semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) measure the representation gap with linear probing and use validation-accuracy increases to explain REPA. Our results show that spatial structure metrics (SSM) provide an additional and more generation-aligned view of the same gap. Figure 17 shows that representation alignment closes the SSM gap between a pretrained encoder such as DINOv2 and the diffusion features, matching the empirical gains observed across models.

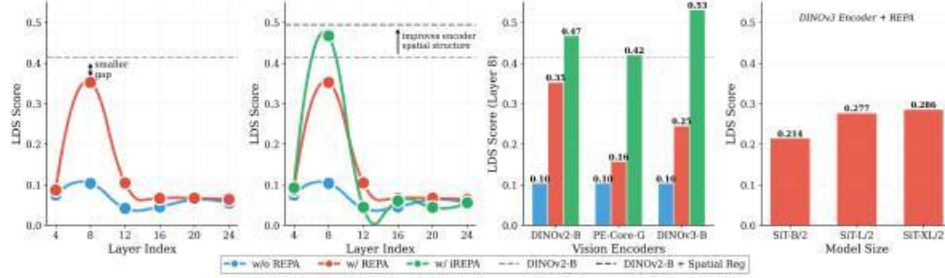


Figure 17: **Representation alignment viewed through the spatial feature gap.** a) The measured gap is between diffusion patch features and vision-encoder patch features, emphasizing the spatial signal that REPA transfers. b) iREPA (§4) deliberately accentuates the vision encoder’s spatial features, trading off some global information to improve generation performance. c) The iREPA-vs-REPA comparison is favorable across different vision encoders. d) The same spatially focused improvement pattern remains visible as model size increases. All reported results use SiT-XL/2 at 400k iterations.

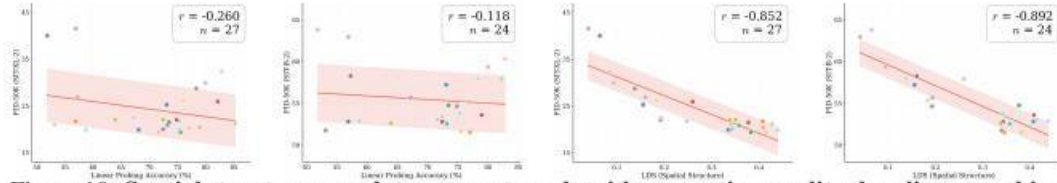


Figure 18: **Spatial structure correlates more strongly with generation quality than linear probing.** Correlation analysis across the same 27 vision encoders under matched REPA evaluation. **Left two:** Linear probing accuracy has weak association with FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$). **Right two:** Spatial structure (LDS) has much stronger association with FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$), giving a clearer quantitative ranking of target encoders. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

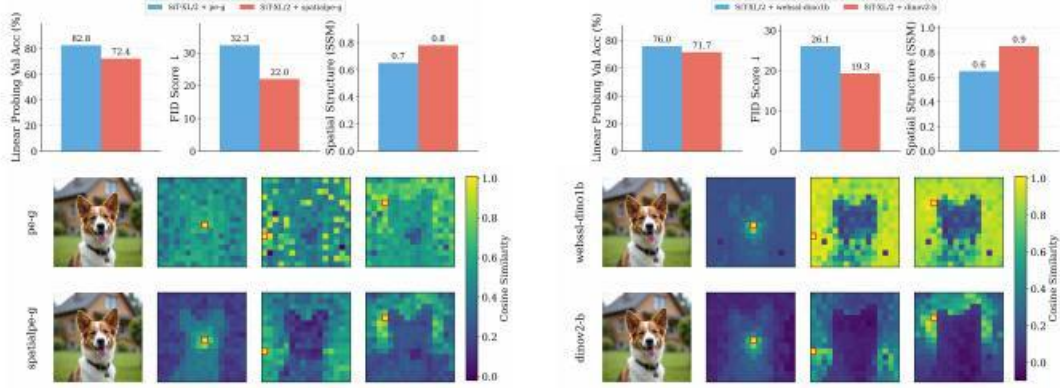


Figure 19: **Motivating encoder comparisons: spatial structure tracks FID better than ImageNet accuracy.** **Top row:** Matched REPA comparisons show that the encoder with higher ImageNet accuracy can have worse generation quality. **Left:** PE-g has higher accuracy than Spatial-PE-g (82.8% vs 72.4%) but worse FID (32.3 vs 22.0), and the same comparison favors Spatial-PE-g on spatial structure (LDS: 0.4 vs 0.1). **Right:** WebSSL-dino1b similarly has higher accuracy than DINOv2-b (76.0% vs 71.7%) but worse FID (26.1 vs 19.3), while DINOv2-b has the stronger LDS score (0.4 vs 0.2). **Bottom row:** Spatial cosine-similarity maps make the metric pattern visible: the lower-FID encoders (Spatial-PE-g, DINOv2-b) preserve clearer spatial coherence than the higher-accuracy alternatives (PE-g, WebSSL).

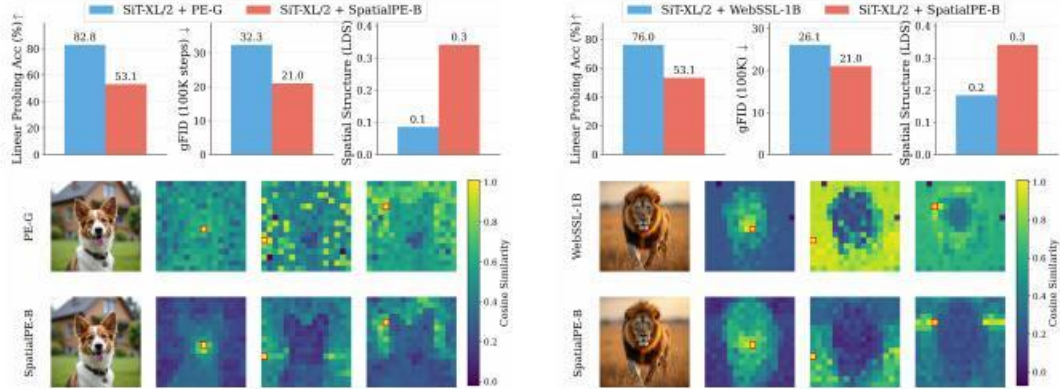


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to PE_{spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Bottom:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize the quantitative mechanism behind spatial normalization by inspecting vision-encoder feature maps before and after normalization. Many encoders contain global overlays that reduce contrast between spatial tokens; for example, tokens in one semantic region can remain highly correlated with unrelated background tokens. Spatial normalization reduces this global component and increases the relative contrast between different spatial locations, which is the spatial property that our correlation analysis links to better generation quality.

The following examples show the same pattern visually: after normalization, feature representations emphasize local spatial structure more clearly than the unnormalized baseline.

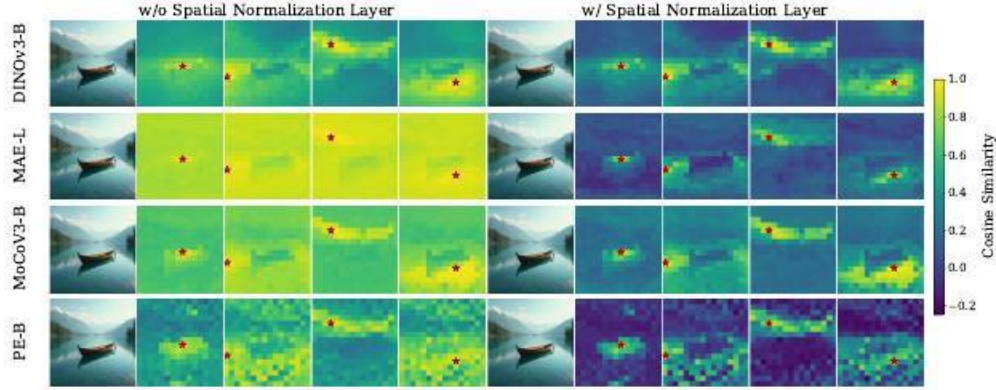


Figure 21: Visualizing impact of spatial normalization (Example 1). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

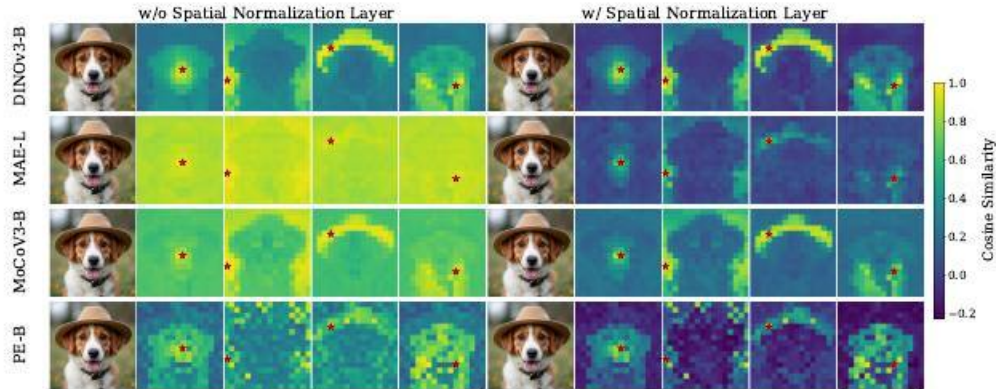


Figure 22: Visualizing impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

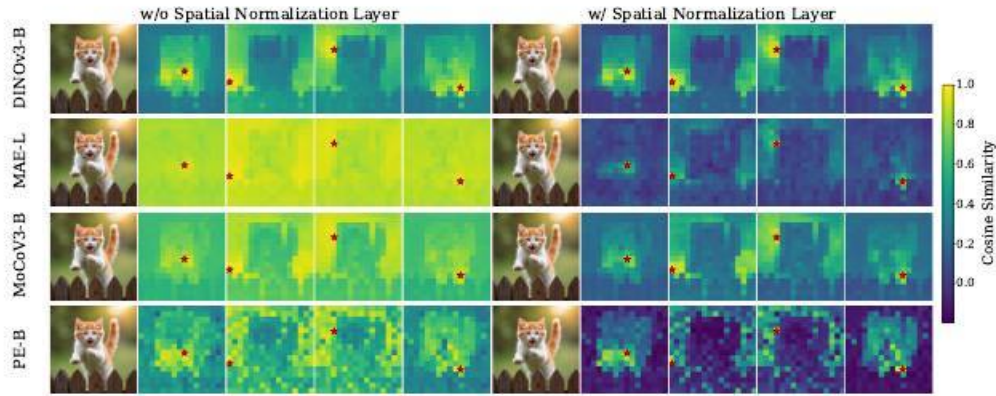


Figure 23: Visualizing impact of spatial normalization (Example 2). Feature similarity maps demonstrate how spatial normalization improves spatial contrast. The original features (left) exhibit a global overlay that reduces distinction between foreground and background regions. After normalization (right), spatial tokens become more locally coherent, with clearer separation between different semantic regions.

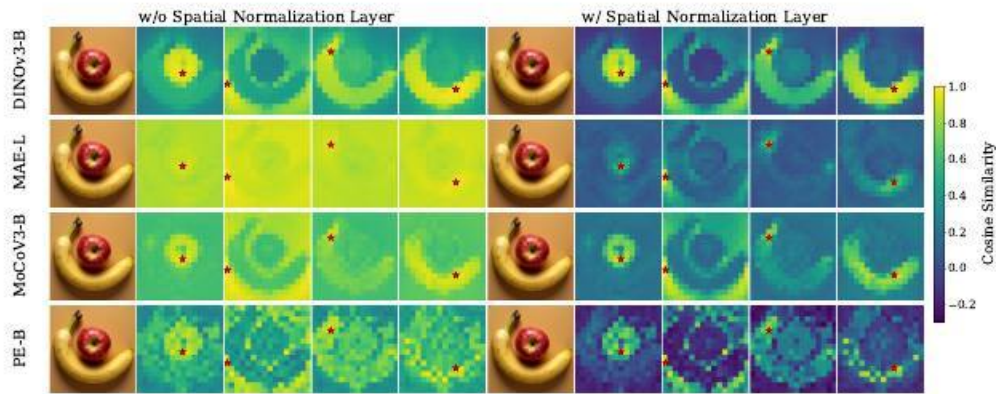


Figure 24: **Visualizing impact of spatial normalization** (Example 4). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We add ImageNet-512 (Deng et al., 2009) experiments to test whether the spatial improvements remain effective at higher resolution. The evaluation follows REPA (Yu et al., 2024) and compares REPA with iREPA across pretrained encoders including DINOv2, DINOv3, WebSSL, and PE. All results use SiT-XL and SiT-B with 50 NFE at inference and no classifier-free guidance. Fig. 25 reports the before/after comparison and shows that iREPA maintains consistent convergence gains at 512x512 resolution across encoder choices.

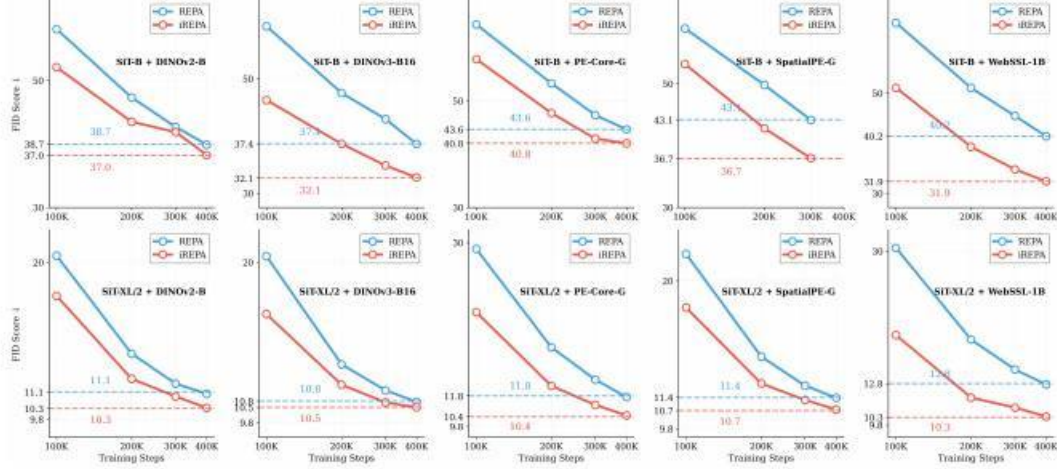


Figure 25: **Convergence results at 512x512 resolution.** The REPA-to-iREPA comparison shows that accentuating spatial features improves convergence at Imagenet 512 as well as Imagenet 256 (Figure 7).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. We also evaluate whether the same spatial modifications transfer beyond class-conditional ImageNet generation to multimodal T2I tasks. Following REPA (Yu et al., 2024), we use MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with pretrained vision encoders including DINOv2, CLIP, WebSSL, and PE. As in REPA (Yu et al., 2024), all models are trained for 150K steps with batch size 256 and evaluated with an ODE sampler using 50 NFE. Fig. 26 compares REPA before and after applying iREPA and shows consistent gains across encoder choices, indicating that the spatial-transfer advantage extends to multimodal T2I generation.

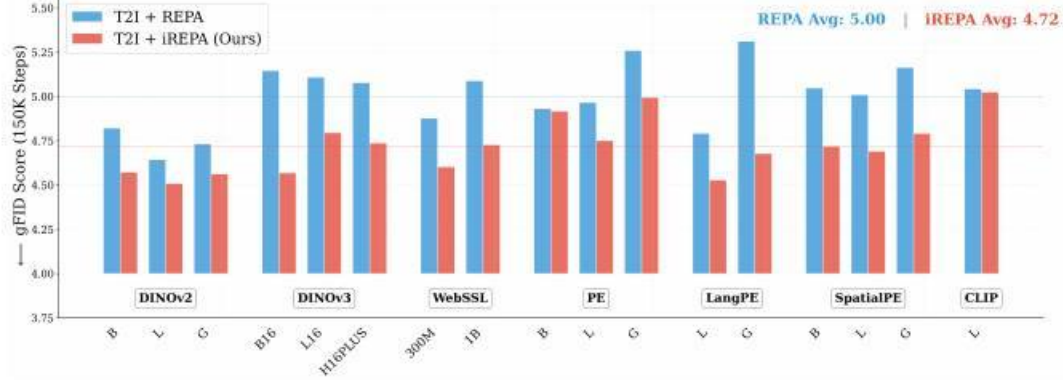


Figure 26: **Text-to-image generation across encoder variants.** The REPA/iREPA comparison shows that transferring spatial features more strongly improves convergence not only on imagenet (refer §4) but also on multimodal T2I tasks. The gains are consistent across pretrained encoders including DINOv2, CLIP, WebSSL, and PE.

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the Section 3 correlation analysis after removing MoCOv3-L and MAE-L, the identified outliers. Figure 28 reports the updated correlations across model sizes. The evidence remains unchanged: spatial structure has much higher correlation with generation performance than linear probing accuracy. After the outliers are removed, linear probing even shows a small positive correlation with gFID, meaning that higher linear-probing performance corresponds to worse generation in this subset. This is consistent with Sec. 2, where targets with higher global semantic performance often yield similar or worse REPA generation performance.

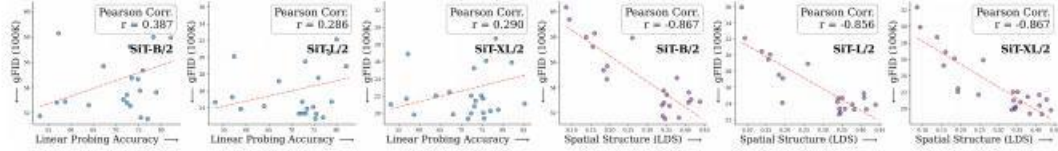


Figure 27: **Correlation analysis without outliers.** Across all model sizes (B, L, XL) spatial structure still shows much higher correlation (Pearson $|r| > 0.85$) with generation performance over linear probing accuracy (Pearson $|r| < 0.38$) after removing the outliers (MoCOv3-L and MAE-L). Interestingly, after removing the outliers linear probing actually shows a small positive correlation with gFID (i.e., as linear probing performance increases, the generation becomes worse). This trend is consistent with the observations discussed in Sec. 2, wherein often target representations with higher global semantic performance (linear probing accuracy) show similar or worse generation performance with representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after application of spatial improvements with REPA. All results are reported using PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs) and with classifier-free guidance scale of 4.0. Similar to quantitative improvements (§4), we also observe that spatial improvements (iREPA) also help improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

In addition to the standard generation metrics (Inception Score, FID, sFID, Precision, Recall), we test whether the evidence remains stable under an alternative quality metric (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). We use CMMD (Jayasumana et al., 2024) with the PyTorch implementation from (Paul, 2024) and the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021). This lets us re-evaluate both the correlation claim from §3 and the iREPA improvement claim from §4 under the same alternative metric.

Spatial structure correlates much higher with generation performance than linear probing. We repeat the large-scale encoder correlation analysis from §3 using CMMD (Jayasumana et al., 2024) instead of gFID. Figure 29 reports SiT-B/2 results at 100K steps with REPA. All spatial metrics retain much higher correlation with CMMD (Pearson $|r| > 0.88$) than linear probing (Pearson $|r| = 0.074$), showing that the main empirical conclusion from §3 is robust to the choice of FID or CMMD.

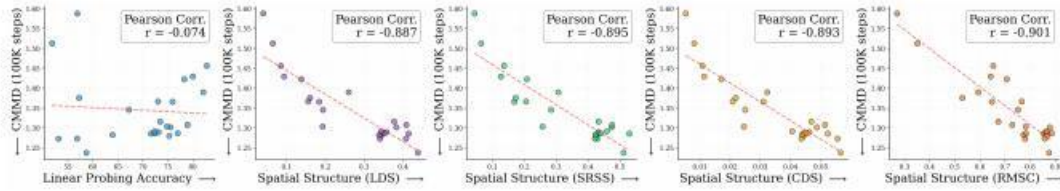


Figure 29: Correlation analysis with CMMD metric. Repeating the §3 correlation analysis with CMMD (Jayasumana et al., 2024) instead of gFID gives the same comparison outcome. For SiT-B/2 at 100K steps with REPA, all spatial metrics correlate much more strongly with CMMD (Pearson $|r| > 0.88$) than linear probing does (Pearson $|r| = 0.074$), preserving the same ranking reported in §3.

Accentuating transfer of spatial features helps improve generation quality. We next evaluate the iREPA-vs-REPA comparison from §4 with CMMD (Jayasumana et al., 2024) alongside the traditional metrics (gFID, sFID, IS, etc.). Table 8 shows that across vision encoders, iREPA improves convergence under both CMMD and the standard metrics (IS, FID, sFID, Prec., Recall). The same direction holds with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** CMMD (Jayasumana et al., 2024) is reported alongside standard metrics for the same REPA-vs-iREPA comparisons. All results use SiT-XL/2, batch size 256, and traditional REPA as the baseline; CMMD is computed with the PyTorch implementation from (Paul, 2024) and the OpenAI ADM reference set (Dhariwal & Nichol, 2021). Across vision encoders, iREPA improves convergence under both CMMD and traditional metrics (IS, FID etc.), showing that the §4 findings are robust to the evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 isolate the effect of spatial normalization (§4) on SAM2-S (46M) encoder features². For SAM2, Table 9 shows improvements from spatial normalization, but the margins are smaller than for encoders with stronger global components. This is expected because spatial normalization removes the mean patch-token component to increase spatial contrast, and SAM2 already has little global information (validation accuracy < 24%), leaving less global overlay to remove (see Figure 30).

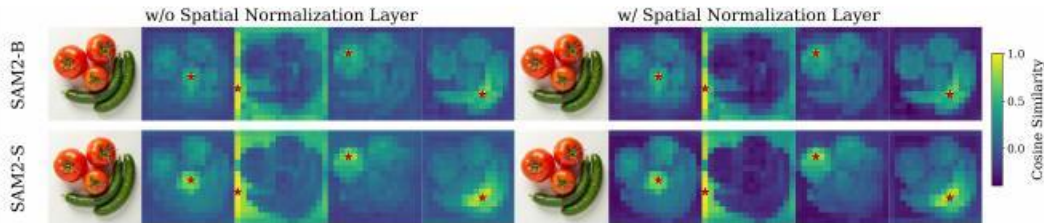


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, spatial normalization improves spatial contrast, but the visible and quantitative margins are smaller than for encoders with stronger global components. This matches the mechanism in §4: spatial normalization removes the mean patch-token component to enhance contrast, while SAM2 already has little to no global information (validation accuracy < 24%). As a result, there is less global signal to remove and the spatial-contrast gain is correspondingly modest.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** Each +iREPA row is compared to the corresponding SAM2-S (46M) REPA baseline at the same training step. All results use SiT-XL/2 without classifier-free guidance.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the §3 correlation analysis using validation accuracy after full-finetuning³ rather than linear probing. Fig. 31 reports SiT-XL/2 results at 100K steps with REPA. For full-finetuning validation accuracy, we finetune each vision encoder with learning rate $5e-5$, warmup ratio 0.1, and 30 total epochs. The conclusion is unchanged: all spatial metrics correlate much more strongly with generation performance than the full-finetuning accuracy baseline. In fact, gFID has a weak positive correlation with full-finetuning validation accuracy (Pearson $r = 0.317$), so higher validation accuracy corresponds to worse generation in this comparison. This matches §2, where targets with higher global semantic performance often produce similar or worse REPA generation quality.

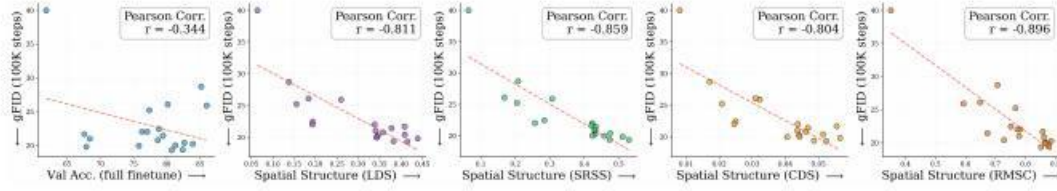


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 with the validation accuracy after full-finetuning instead of linear probing. All results are reported using SiT-XL/2 (100K steps) and REPA. All spatial metrics show much higher correlation with generation performance than full-finetuning accuracy. Interestingly, gFID actually shows a weak positive correlation with the validation accuracy after full-finetuning (Pearson $r = 0.317$), i.e., as validation accuracy increases, the gfid increases, and generation performance becomes worse. This is consistent with the observations discussed in §2, wherein often target representations with higher global semantic performance show similar or worse generation performance with representation alignment (REPA).

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** When combined with JiT (Li & He, 2025a), iREPA compares favorably with recent pixel-space diffusion methods such as DeCo (Ma et al., 2025b). All results use 200K training iterations, batch size 256, and 50-step Euler sampling without classifier-free guidance. DeCo results are taken directly from (Ma et al., 2025b).